(5) $y(t) = 10x(2t)^{12}$

a). $x_1 \to y_1$, $x_2 \to y_2$

$y_1(t) + y_2(t) = 10x_1(2t)^{12} + 10x_2(2t)^{12}$

$(y_1(t) + y_2(t) = 10(x_1(2t)^{12} + x_2(2t)^{12})$

$= 10x_1(2t)^{12} + 10x_2(2t)^{12}$

∴ the equation passes the superposition test

$ax_1 \to ay_1$

$y(ax(t)) = 10(ax(2t)^{12}) = 10ax(2t)^{12}$ ↳ matches

$ay = a(10x(2t)^{12}) = 10ax(2t)^{12}$

∴ The equation is homogenious

∴ The System is linear

b) $y(t) = 10x(2t)^{12}$ $\quad x(t-t_0) \to y(t_0-t_0)$

$y(t-t_0) = 10x(2t-2t_0)^{12}$ ↳ does not match

for $x(t-t_0)$ $\quad y(t) = 10x(2t-t_0)^{12}$

∴ the system is time variant

c) we tested the system by comparing $x(t_n - t_0)$ to $y(t - t_0)$ to a system where t_0 is subtracted from t in the equation. ~~Because This proved~~ The two systems did not match, proving they were time variant. Thus, the result of the system depends on time as a factor.